

Hackathon Problem Statements

HealthTech

	Themes	Problem Statement
01	HealthTech	AI-Based Early Disease Detection System: In many cases, diseases such as heart conditions, diabetes, and respiratory disorders are diagnosed at later stages due to lack of early screening tools. This delay leads to increased healthcare costs and reduced survival rates. Proposed Solution: Develop an AI/ML-based system that analyzes user inputs such as symptoms, medical history, and lifestyle data to predict the probability of diseases at an early stage. The system should provide risk scores, suggest preventive measures, and recommend whether medical consultation is required.
02	HealthTech	Remote Patient Monitoring System for Rural Areas: In rural and semi-urban areas, continuous monitoring of patients is difficult due to lack of infrastructure and healthcare professionals. Patients often need to travel frequently for basic check-ups. Proposed Solution: Design an IoT-based system that collects patient health data (heart rate, temperature, oxygen levels) and sends it to a centralized platform accessible by doctors. The system should generate alerts in case of abnormal readings and allow remote monitoring.
03	HealthTech	Smart Electronic Health Record (EHR) System: Patient data is often scattered across multiple hospitals, making it difficult for doctors to access complete medical history. This leads to repeated tests and inefficient treatment. Proposed Solution: Develop a secure digital platform where patient health records are stored and can be accessed by authorized medical professionals. The system should include authentication, encryption, and easy sharing features.
04	HealthTech	Automated Mental Health Assessment Platform: Mental health issues are rising, but early detection remains a challenge due to lack of awareness and hesitation in seeking help. Proposed Solution: Build an application that can assess mental health status using questionnaires, behavior tracking, and AI-based analysis. The system should suggest basic coping strategies and recommend professional help when required.

05	HealthTech	Medicine Adherence and Reminder System: Patients often forget to take medicines on time, leading to ineffective treatment and health complications. Proposed Solution: Create a smart application that reminds users about medication schedules, tracks adherence, and sends alerts to caregivers in case of missed doses. The system should generate reports for doctors and family members.
06	HealthTech	AI-Based Medical Report Analysis System: Medical reports are complex and difficult for patients to understand, leading to confusion and misinterpretation. Proposed Solution: Develop a system that extracts information from medical reports (PDF/image) and explains it in simple language. The system should highlight abnormal values and provide basic health insights.
07	HealthTech	Telemedicine Platform for Remote Consultation: Access to healthcare professionals is limited in many regions, especially rural areas. Patients face delays in getting medical advice. Proposed Solution: Design a telemedicine platform that allows patients to consult doctors via video/audio calls. The platform should support appointment booking, digital prescriptions, and low-bandwidth communication.
08	HealthTech	Affordable Health Tracking and Fitness Monitoring System: Many individuals do not regularly track their health due to lack of affordable solutions. Proposed Solution: Develop a cost-effective health tracking system that monitors daily activities, fitness levels, and basic health parameters. The system should provide personalized recommendations for improving health.
09	HealthTech	Emergency Response and Ambulance Management System: Delayed ambulance response and poor coordination between hospitals lead to critical loss of time during emergencies. Proposed Solution: Create a system that tracks ambulance locations, suggests the fastest route using real-time traffic data, and alerts hospitals in advance about incoming patients.

10	HealthTech	Fake Medicine Detection and Verification System: The circulation of counterfeit medicines poses serious health risks to patients. Proposed Solution: Develop a system that verifies the authenticity of medicines using barcode/QR scanning. The system should connect to a centralized database and alert users if the medicine is suspected to be fake.
11	HealthTech	AI-Based Symptom Checker and Triage System: Patients often struggle to understand the severity of their symptoms and whether immediate medical attention is required. Proposed Solution: Build an AI-based system where users can input symptoms and receive possible diagnoses along with urgency levels (low, medium, high). The system should guide users on next steps.
AgriTech		
12	AgriTech	AI-Based Crop Disease Detection System: Farmers often fail to identify crop diseases at an early stage, leading to reduced yield and financial losses. Lack of expert guidance in rural areas worsens the situation. Proposed Solution: Develop an AI-powered system that uses image processing to detect crop diseases from leaf images. The system should provide disease identification, severity level, treatment suggestions, and preventive measures in regional languages.
13	AgriTech	Smart Irrigation Management System: Excessive or insufficient irrigation leads to water wastage and poor crop growth. Farmers lack real-time insights into soil moisture and weather conditions. Proposed Solution: Design an IoT-based irrigation system that monitors soil moisture, weather forecasts, and crop type to automate irrigation. The system should optimize water usage and provide alerts for manual intervention when required.
14	AgriTech	Precision Farming Decision Support System: Farmers rely on traditional practices instead of data-driven decisions, resulting in inefficient use of fertilizers, pesticides, and resources. Proposed Solution: Develop a platform that integrates satellite data, soil health data, and weather

		forecasts to provide personalized recommendations on crop selection, fertilizer usage, and sowing time.
15	AgriTech	Farmer-to-Market Direct Trading Platform : Farmers often depend on intermediaries, reducing their profit margins and market transparency. Proposed Solution: Create a digital marketplace where farmers can directly connect with buyers, retailers, or consumers. The platform should include price comparison, demand forecasting, and logistics support.
16	AgriTech	Crop Yield Prediction System: Uncertainty in crop yield affects financial planning and supply chain management. Proposed Solution: Build an AI/ML model that predicts crop yield based on historical data, weather conditions, soil quality, and farming practices. The system should assist farmers and policymakers in planning and decision-making.
17	AgriTech	Smart Pest Monitoring and Control System: Pest infestations cause significant crop damage, and timely detection is difficult without continuous monitoring. Proposed Solution: Develop an IoT and AI-based pest monitoring system using sensors and smart traps. The system should detect pest activity, predict outbreaks, and recommend targeted pesticide usage.
18	AgriTech	Weather-Adaptive Farming Advisory System: Unpredictable weather patterns due to climate change impact crop productivity and farmer income. Proposed Solution: Create a system that provides real-time weather-based advisories, including alerts for extreme conditions and suggestions for adaptive farming practices.
19	AgriTech	Smart Cold Storage and Inventory Management System: Post-harvest losses occur due to inadequate storage facilities and poor inventory management. Proposed Solution: Design a smart cold storage system with IoT sensors to monitor temperature, humidity, and inventory. The system should alert users about spoilage risks and optimize storage conditions.

20	AgriTech	Smart Greenhouse Automation System: Manual greenhouse management leads to inefficient control of temperature, humidity, and light conditions. Proposed Solution: Develop an automated greenhouse system using IoT and AI to regulate environmental parameters for optimal crop growth.
21	AgriTech	Agricultural Waste Management and Recycling System: Improper disposal of agricultural waste leads to environmental pollution and loss of potential resources. Proposed Solution: Design a system that converts agricultural waste into useful products like biofuel, compost, or animal feed. The platform should guide farmers on sustainable waste management practices.
FINTECH		
22	FinTech	UPI Fraud Intelligence & Behavioral Biometrics System: UPI frauds are increasing, and OTP/PIN-based systems are no longer sufficient to ensure security. Proposed Solution: Design a system that uses behavioral biometrics (typing speed, touch patterns, transaction habits) to detect anomalies in real-time and prevent fraudulent transactions before completion.
23	FinTech	Decentralized Identity (DID) for Financial Services: Users repeatedly share sensitive personal data across platforms, increasing the risk of data breaches. Proposed Solution: Build a blockchain-based decentralized identity system where users control their KYC data and grant temporary, secure access to financial institutions when needed.
24	FinTech	AI-Based Scam & Phishing Detection Assistant: Users frequently fall victim to financial scams via SMS, calls, and fake links due to lack of real-time verification tools. Proposed Solution: Develop an AI assistant that scans messages, links, and calls in real-time to detect potential scams and warn users instantly with risk scores and explanations.

25	FinTech	Smart Contract-Based Insurance Claim System: Insurance claim processes are slow, manual, and prone to fraud. Proposed Solution: Create a smart contract-based system where claims are automatically verified and processed based on predefined conditions and real-time data inputs.
26	FinTech	Dynamic Credit Line System (Real-Time Adaptive Lending): Traditional lending systems assign fixed credit limits based on static historical data, which does not reflect a user's current financial condition. This leads to inefficient credit utilization and increased default risks. Proposed Solution: Build a dynamic credit system that continuously evaluates user financial behavior and adjusts credit limits in real time. The system should: • Analyze income patterns, transaction behavior, and repayment history • Use AI models to assess creditworthiness dynamically • Adjust credit limits based on real-time risk assessment • Provide transparency in credit decisions and recommendations for improvement
27	FinTech	Green Finance & Carbon Footprint Tracking Platform: There is limited awareness and accessibility for individuals to track and offset their carbon footprint through financial actions. Proposed Solution: Develop a platform that links financial transactions with environmental impact. The system should: • Estimate carbon footprint based on spending patterns • Suggest eco-friendly alternatives • Provide carbon credits or rewards for sustainable behavior • Enable green investments
28	FinTech	Smart Bill Reminder & Auto-Payment Optimization System: Users often miss bill payments or pay late fees due to poor tracking. Proposed Solution: Build a system that: • Tracks recurring bills (electricity, rent, subscriptions) • Sends reminders before due dates • Suggests optimal payment timing • Enables auto-pay with safety checks

29	FinTech	QR-Based Offline Payment Logging System: Small vendors in low-network areas struggle with digital payment tracking. Proposed Solution: Design a system that: • Enables offline QR-based transaction logging • Syncs data when internet is available • Maintains transaction history • Generates simple financial reports
30	FinTech	AI-Based Expense Categorization & Smart Insights System: Users make hundreds of digital transactions monthly but lack clear insights into where their money is being spent. Manual tracking is tedious and inaccurate. Proposed Solution: Develop an AI-based system that automatically categorizes transactions (food, travel, bills, etc.) using NLP and pattern recognition. The system should: • Provide visual dashboards of spending habits • Detect unusual spending patterns • Suggest cost-cutting opportunities • Generate monthly financial summaries
31	FinTech	UPI-Based Split Expense & Group Settlement Platform: Managing shared expenses among friends, roommates, or teams is often confusing and leads to delayed settlements. Proposed Solution: Build a smart expense-sharing platform integrated with UPI that: • Tracks group expenses in real time • Automatically calculates who owes whom • Suggests optimal settlement paths (min transactions) • Sends reminders for pending payments
OPEN INNOVATION		

	Open Innovation	The Open Innovation Track is designed to give you complete freedom to think, create, and innovate beyond predefined problem statements. This is your opportunity to identify real-world challenges and develop impactful, technology-driven solutions. In this track, you are encouraged to explore ideas across any domain such as healthcare, agriculture, fintech, education, environment, smart cities, and more. The focus is on solving meaningful problems that can create a positive impact on society. You Can Use Technologies Like:
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